

Guardian Band

WEARABLE FALL DETECTION & EMERGENCY ALERT SYSTEM for ELDERLY CARE

* PROBLEM STATEMENT:-

Elderly people living alone may not receive immediate help after falls, fainting or medical emergencies.

Delayed assistance can increase the risk of severe injury or death.

The aim of this project is to develop a wearable device capable of detecting abnormal fall or fainting events or prolonged inactivity & automatically notify caregivers or family members through SMS alerts.

* OBJECTIVE:-

To design a wearable biomedical safety device using Arduino UNO & motion sensors that can:-

- 1] Detect fall, fainting or abnormal body posture
- 2] Monitor inactivity after a fall
- 3] Trigger an emergency alert
- 4] Send SMS notification to predefined contacts.
- 5] Reduce delay in emergency response
- 6] Minimize false alarm using multiconditioned analysis

* REAL WORLD USE CASE *

- 1] Elderly people living alone
- 2] Patients with Balance Disorder
- 3] Stroke risk Patients
- 4] Infants under Supervision
- 5] Home healthcare Monitor.

* COMPONENTS REQUIRED *

- 1] ARDUINO UNO :- main controller
Processes sensor data
- 2] MPU 6050 SENSOR :- detects motion & orientation.
Contains accelerometer & gyroscope
- 3] SIM 900A GSM MODULE :- sends emergency SMS alert
- 4] BUZZER :- Provides audible warning before SMS
- 5] PUSH BUTTON :- Cancels false alarm.
- 6] LEDs :- Indicate System status
- 7] BREADBOARD & JUMPER WIRES :- circuit Connection

★ WHAT IS ARDUINO UNO ?

It's a microcontroller board used for electronics prototyping & embedded system development.

It receives sensor input, processes logic & controls output devices. In this project, it acts as the brain of the system.

★ WHAT IS MPU 6050?

It's a motion tracking sensor containing:-

- (1) Accelerometer :- Detects acceleration & movement
- (2) Gyroscope:- Detect rotation & orientation changes

This sensor is used to identify abnormal movement patterns associated with falls or collapse.

★ HOW DETECTION WORKS?

The system detects possible emergencies using:-

- 1] Sudden acceleration
- 2] Significant orientation change
- 3] Slow abnormal posture change
- 4] Prolonged inactivity after abnormal posture change

The system can identify:-

- 1] Hard falls
- 2] Slipping incidents
- 3] Fainting
- 4] Slow Collapse events

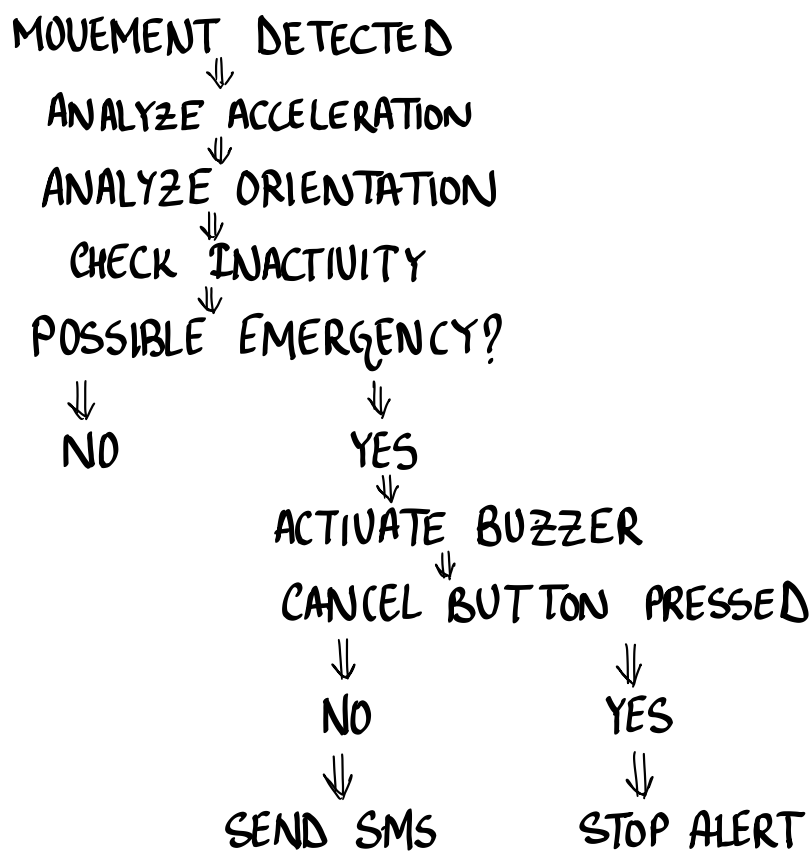
If these conditions occur together, the system classifies the event as a possible medical emergency.

★ FALSE ALARM PREVENTION:-

To avoid false alarm caused by normal movements such as sitting or jumping, the system includes:-

- 1] Multi-Condition detection
- 2] Inactivity monitoring
- 3] Recovery Observation
- 4] Manual Cancel button.

★ SYSTEM FLOW:-



★ FUTURE IMPROVEMENTS:-

- 1] GPS Location Tracking
- 2] Pulse monitoring to improve fainting & unconsciousness detection accuracy.
- 3] Mobile application integration
- 4] Cloud Based emergency monitoring
- 5] AI based fall classification

* NORMAL v/s ABNORMAL HUMAN MOVEMENT ANALYSIS:-

The system must differentiate between normal daily activities & possible medical emergencies:-

NORMAL ACTIVITIES:- Walking
Sitting
Standing
Lying down intentionally
Hand movement

ABNORMAL ACTIVITIES:- Sudden hard falls
Slipping incidents
Fainting
Collapse
Prolonged inactivity after abnormal posture change

The detection system therefore analyzes:-

- ① Acceleration
- ② Orientation
- ③ Inactivity duration
- ④ Recovery Movement

* CHALLENGES:-

One of the major challenges in wearable fall detection system is differentiating between normal daily activities & actual medical emergencies

Normal activities like jumping, sitting suddenly or dropping the device may appear similar to fall

Hence, to improve reliability, the system combines:-

- | | |
|---------------------------|-------------------------|
| 1] Motion analysis | 3] inactivity tracking |
| 2] Orientation monitoring | 4] recovery observation |